

Review

Data-Driven Leadership in Higher Education: Advancing Sustainable Development Goals and Inclusive Transformation

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Abstract: The transformative function of data-driven leadership in higher education institutions (HEIs) is becoming crucial for advancing sustainable development. By integrating data-driven decision-making with Sustainable Development Goals (SDGs), particularly SDG4 (quality education) and SDG10 (reduced inequalities), HEIs can improve the efficacy, inclusivity, and employability of their graduates. To examine this influence, this study implements a systematic literature review (SLR) that adheres to the PRISMA standards and integrates empirical and theoretical insights regarding data-driven leadership in HEI governance, teaching, and learning strategies. The results indicate that combining data analytics into decision-making processes improves institutional efficacy, aligns curricula with the market demands, strengthens student outcomes, and cultivates an inclusive and sustainable academic environment. Moreover, this study introduces a conceptual model connecting sustainable development and data-driven decision-making, offering a structured framework for HEIs to navigate digital transformation responsibly. In addition, this model also emphasizes the importance of balancing technology, ethics, and human-centric leadership in developing educational institutions that are prepared for the future. Ultimately, these insights provide practical advice for academic leaders and policymakers aligning HEI strategies with global sustainability objectives. By advocating for innovative, inclusive, and data-driven leadership, HEIs can promote long-term societal transformation and higher education excellence.



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Keywords: data-driven leadership; higher education institutions; Sustainable Development Goals; transformative education; diversity, equity, inclusion, equality; employment; teaching and learning strategies

1. Introduction

1.1. Background

Higher education institutions (HEIs) confront several problems in the twenty-first century, as economic, environmental, demographic, political, labor market, technical, social, and health concerns are changing quickly [1–4]. The twenty-first century is a period of transition, driven by many influences and the necessity of determining our societies' futures [1]. Moreover, the importance of education as a primary contributor to a sustainable future is omnipresent [5–14]. Therefore, if education is to play the central role it can and should play in propelling society and humanity forward, then the time to rewire HEIs is now.

While there is much study on data-driven leadership and decision-making in education, certain aspects, especially in the context of sustainable development and comprehensive transformation of HEIs, have not been well investigated. Specifically, the role of

data analysis in graduate employability, teaching methodologies, diversity, equality, and inclusion initiatives are usually brushed over in the rhetoric of transformation in HEIs and their alignment with sustainable development. For instance, Wilcox et al. argue that there is limited empirical data to support the complete inclusion for all students and much less knowledge on the function of data-based decision-making in inclusive education [15]. This implies that educational data mining is still not a complete area [16], even though empirical studies have found that data can accelerate global innovation and improvement in education [14,17–26]. As a result, we conducted a systematic literature review (SLR) to investigate and explore how data-driven leadership and decision-making may lead to effective educational reform and assure long-term growth. To attain this objective, it was critical to comprehend the transformation in the literature on HEIs, focusing on graduate-level employability, teaching and learning strategies, diversity, equality, and inclusion from a global perspective. Thus, an SLR was selected to offer a comprehensive synthesis of the current research, providing a broad overview of the current knowledge. Additionally, it ensures transparency and replicability by adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) standards and following a structured literature search, selection, and analysis process. Furthermore, an SLR identifies absences in the literature, which informs higher education management policies and practices and guides future research.

Sustainability and data-driven decision-making in higher education [27–31] are gaining importance due to global concerns such as diversity, climate change, technological advancements, equity, and shifting labor market demands. As a result, this study is pertinent. This pressing need drives the research to explore how HEIs can leverage data-driven leadership to address challenges and promote sustainable development. One distinguishing feature of this study is its comprehensive analysis of integrating data-driven leadership methods into the sustainability and transformation efforts of HEIs. By examining how data-driven leadership and decision-making can facilitate successful educational reform and secure sustained expansion in HEIs, this research introduces a novel element to the body of knowledge. The results of our study demonstrate the importance of data-driven leadership in promoting diversity, equity, and inclusion (DEI) in HEIs, as well as augmenting the employability of graduates and teaching and learning strategies. Such insights establish the foundation for future research and strategic planning regarding the global transition of HEIs. Ultimately, this review is meant to provide a broad overview of the current state of transformation in tertiary institutions. Thus, this research aims to serve as a tool for HEIs to make informed decisions and develop strategies to position them at the vanguard of the global HEIs transition. This study can be used to inform future research by laying the groundwork for expanding and validating hypotheses in a variety of related fields. Our findings could also be used to spark introspection, debate, and innovative planning, as organizations develop their approaches to facilitating future transformation. By leveraging this research, HEIs, policymakers, and researchers can work together to create a more promising future.

1.2. Literature Review

Through research, knowledge dissemination, and skill development, HEIs are instrumental in developing sustainable societies. HEIs must adapt strategically to address the intensification of global challenges related to climate change, labor market disruptions, and digitalization, while also aligning with sustainable development frameworks. Scholars contend that universities worldwide are acknowledging their obligation to provide students and communities with the necessary tools to actively engage in carbon neutrality initiatives, climate change mitigation, and adaptation [32–34]. HEIs are critical contributors

to sustainability, mainly through the implementation of curricular reforms, research, and institutional policies that foster environmental awareness. However, incorporating sustainability in HEIs remains fragmented, with varying levels of institutional commitment and policy implementation across regions [35].

In addition to environmental concerns, HEIs must adapt to fluctuations in the labor market and the economy. Researchers have emphasized the growing misalignment between academic training and labor market requirements, resulting in a skills imbalance that impedes the employability of graduates [36,37]. Machin and McNally underscore the necessity of competency-based education models that facilitate the seamless integration of graduates into the workforce [38]. Despite these apprehensions, numerous universities implement conventional curricula, which may not adequately equip students for the swiftly changing job market. The emergence of digitalization and technological advancements has further disrupted HEIs, necessitating the adaptation of their pedagogical approaches. Studies conducted by Chigbu et al. and Fülöp et al. indicate that digital transformation in HEIs improves teaching and learning outcomes [39,40]. However, numerous institutions cannot capitalize on emergent technologies due to lacking infrastructure and faculty capacity. The digital divide in global higher education remains a significant issue, restricting the availability of inclusive and technology-driven education in specific regions.

The Sustainable Development Goals (SDGs) offer universities a global framework to ensure that their policies are in accordance with sustainability, inclusion, and employability. The implementation of SDG4 (Quality Education) and SDG10 (Reduced Inequalities) in HEIs promotes inclusive and equitable education, guaranteeing that all individuals have access to learning opportunities regardless of their gender, socioeconomic status, ethnicity, or disability status [35,41,42]. Žalėnienė and Pereira found that integrating sustainability into HEI curricula and institutional strategies cultivates a culture of sustainability that transcends university environments [35]. Nevertheless, policy inconsistencies, funding constraints, and faculty resistance present substantial obstacles to integrating SDGs [43]. Additionally, there is a dearth of systematic evaluation to assess the impact of these initiatives, although HEIs are increasingly integrating sustainability-oriented curricula and research. Adams et al. underscore that HEIs implementing SDG frameworks are more likely to experience improved stakeholder collaboration and community engagement, which are essential for accomplishing long-term sustainability goals [43]. Nevertheless, there is still a lack of empirical data regarding how HEIs can integrate the SDGs into their governance, curriculum, and administrative structures. Understanding how universities can transition from policy commitments to measurable impact in areas such as graduate employment outcomes, industry partnerships, and social equity is a significant research gap.

Ensuring DEI across universities worldwide is one of the primary obstacles to higher education transformation. Research suggests that numerous leading universities worldwide continue to encounter challenges regarding gender, ethnicity, and socioeconomic diversity [44,45]. Institutional biases, systemic discrimination, and financial barriers persist in excluding marginalized populations from quality higher education opportunities despite the introduction of policy initiatives to promote equity in access to education. The absence of data-driven decision-making that is DEI-focused further complicates the process of establishing equitable learning environments. Existing research indicates that, although universities collect extensive institutional data on student demographics, enrolment trends, and academic performance, few institutions effectively integrate these data to implement targeted interventions that foster inclusivity. Consequently, this study addresses this disparity by investigating how HEIs can utilize data analytics to develop sustainable DEI strategies and combat inequities.

Data-driven leadership is becoming more widely acknowledged as critical in transforming HEIs. It enables institutional leaders to make informed, evidence-based decisions regarding academic policies, resource allocation, and student success initiatives [27,29]. Despite its increasing prevalence, empirical research on data-driven leadership's effectiveness, ethical considerations, and sustainability is still lacking. Numerous HEIs cannot establish comprehensive data-driven policies due to inadequate technical infrastructure, faculty expertise, and strategic frameworks [26]. Additionally, to prevent unintended biases in decision-making, it is imperative to address data privacy concerns and ethical dilemmas associated with learning analytics and AI-driven education technologies [28]. By synthesizing current research and proposing a conceptual model that incorporates data-driven leadership, sustainability, and institutional governance, this study contributes to filling these gaps.

1.3. Conceptual Model for the Role of Data-Driven Leadership in Sustainable HEIs Transformation

This study uses a conceptual framework to investigate the role of data-driven leadership in the long-term transformation of HEIs. This framework serves as both an analytical tool and a model for elucidating causal links, offering a systematic way to understand how data-driven leadership promotes the long-term transformation of HEIs. Illustrated in Figure 1 is the model that encompasses the framework's core elements.

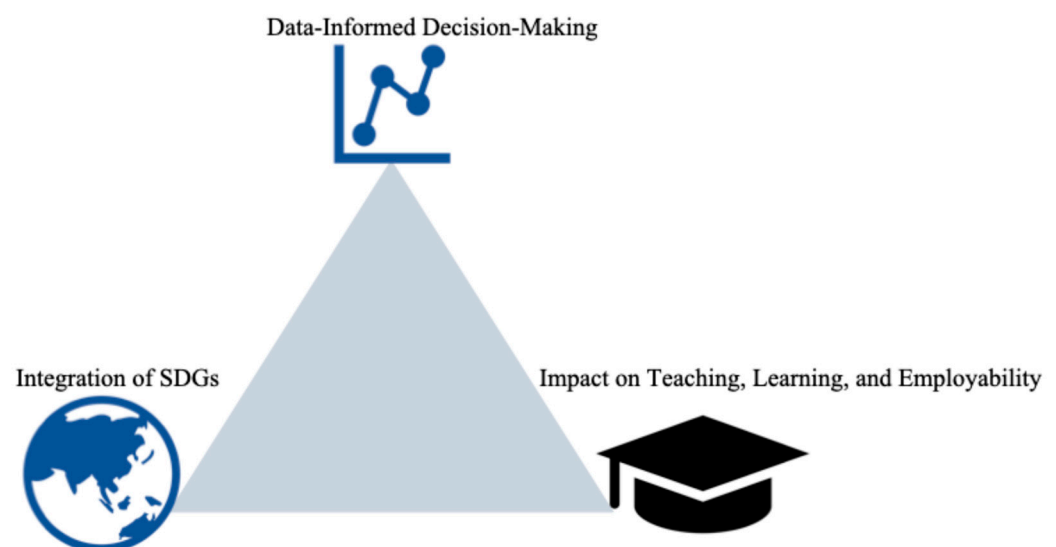


Figure 1. Data-driven leadership in HEIs.

The conceptual approach emphasizes three significant components: data-driven decision-making, SDG integration, and the impact on teaching, learning, and employment. Data-informed decision-making is how higher education leaders acquire, evaluate, and interpret data to make strategic decisions. Studies have shown that data collection and analysis contribute to informed strategic decisions that enable institutional transformation, assuring evidence-based judgments and mitigating the risks associated with intuition-based decision-making [27,46,47]. This approach leads to more effective policies, optimal resource allocation, and focused interventions that promote institutional goals. SDG integration entails connecting higher education policies and activities with the SDGs, including SDG4 and SDG10. When HEIs integrate their strategy with the SDGs, they develop a framework that drives their transformational activities, ensuring that the institution's every action contributes to global sustainability goals. This connection encourages sustainability in educational methods, increases the institution's social responsibility, and boosts its reputation and outreach. The impact on teaching, learning, and employability investigates how

data-driven leadership influences teaching approaches, learning outcomes, and graduate employability. Data-driven decisions result in better teaching strategies, which increase learning outcomes and graduate employability. The continuous feedback loop from data analysis enables incremental enhancements in these areas, resulting in improved teaching methods, better learning outcomes, and higher graduate employability rates, contributing to the institution's overall success and ability to attract and retain students.

The model's transformational techniques include incorporating SDGs into HEI core operations such as teaching methods, research initiatives, and community collaborations [48–50]. Giesenbauer and Müller-Christ illustrate how linking HEI operations with SDGs fosters institutional success and broader societal impact [51]. This integration drives the institution's strategic direction, ensuring that transformational projects are successful and sustainable and promoting a comprehensive education and institutional leadership approach. The benefits include sustainable educational methods, improved research outputs relevant to global needs, and more significant community relationships. The iterative continuous improvement process consists of a continual feedback loop that examines and reviews outcomes utilizing data-driven techniques. Feldman and Czerniewicz highlight the importance of constant feedback loops from data analysis to inform teaching strategies [52], which improves overall educational quality and student employability. Regular feedback ensures the institution can adapt its tactics to changing situations and new insights, which is critical for progressing toward long-term change. This flexibility produces a dynamic and responsive institution that can continuously develop and innovate. By including this model throughout the study, the framework acts as a primary instrument for understanding and describing the long-term transformation of HEIs through data-driven leadership.

The remaining sections of this paper are organized as follows: methods and materials are discussed in Section 2, which primarily focuses on the SLR data collection technique. The findings are discussed in Section 3. Section 4 concludes the study.

2. Materials and Methods

This study employs a narrative systematic literature review (SLR) to examine the role of data-driven leadership in the long-term transformation of HEIs. The narrative SLR methodology was chosen to systematically collect, analyze, and synthesize existing research. The three main goals of SLR are as follows: (1) generating or evaluating hypotheses about how or why events occur; (2) answering problems that individual investigations could not address; and (3) synthesizing the state of knowledge in an area, from which future research priorities may be established [53]. This review accomplishes these goals. This SLR generates new hypotheses regarding the influence of data-driven leadership on sustainable development in HEIs by examining various studies. For instance, it posits that the effective integration of data-driven decision-making processes into leadership strategies can substantially improve the employability of graduates by identifying skill deficiencies and aligning curricula with market demands. A study by Chigbu et al. indicates that HEIs can enhance student satisfaction and improve learning outcomes by customizing their teaching methods through data analytics to monitor student performance and feedback [39]. This review integrates findings from various sources to address intricate issues that individual studies may not resolve. In particular, it investigates the potential of data-driven leadership to concurrently impact multiple aspects of the HEI transformation, including diversity, equity, inclusion, and alignment with the SDGs. This approach provides a comprehensive understanding that is not present in individual studies. For example, the impact of data-driven decision-making on the employability of graduates or teaching strategies may be the focus of individual studies. By integrating these perspectives, this SLR demonstrates how a comprehensive data-driven approach can simultaneously address both areas, resulting in

more integrated and effective educational reforms. Following an SLR is motivated by the fact that each topic is based on a methodology that contains a set of transparent methods and synthesis components from communal repositories of reputable existing research, reasonings, and analyses [39].

This study complied with the PRISMA guidelines and employed a systematic literature search, selection, and analysis methodology. The PRISMA flow chart in Figure 2 demonstrates total openness and accuracy of why the SLR was performed, how papers were located and chosen, and what we discovered. The diagram depicts how the studies were selected, from the first database search to the final included studies. Prior research has verified PRISMA efficacy, thereby bolstering the credibility and dependability of evaluations [39,54–57].

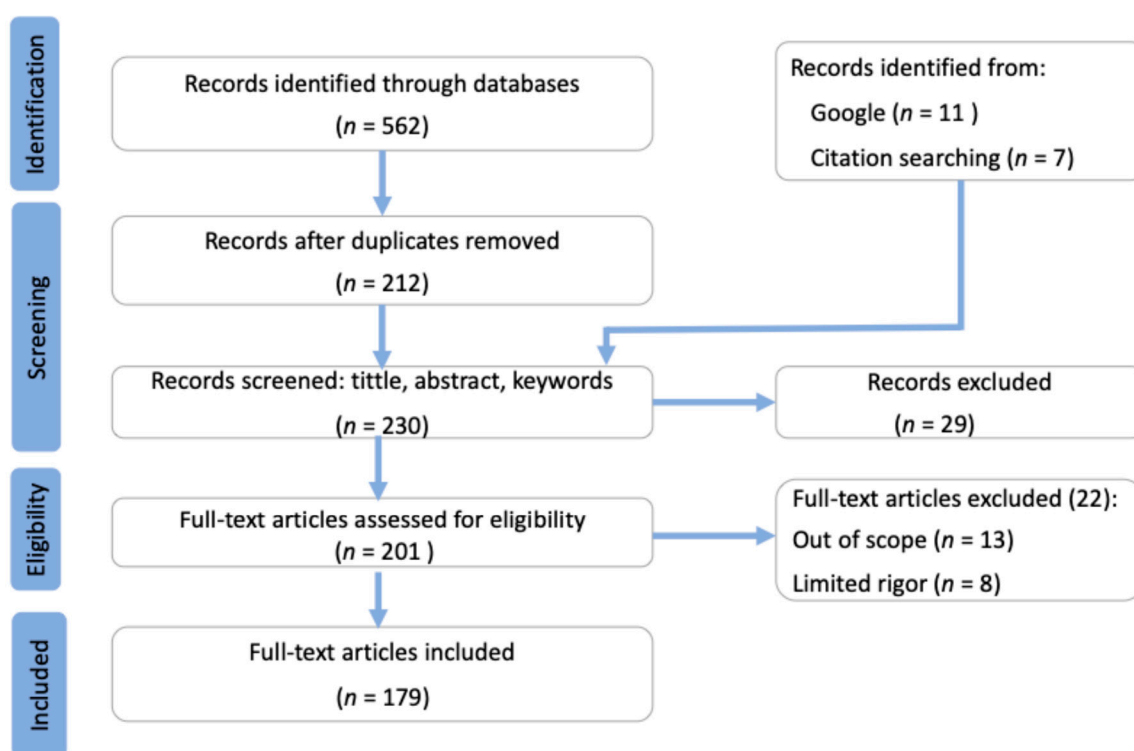


Figure 2. Screening process of SLR. Flow diagram in accordance with PRISMA.

The initial search was conducted for published journal articles using the search words “transformation in HEIs”, “data-driven leadership”, “education and sustainable development”, “employability”, “teaching and learning”, and “diversity, equity, and inclusion” on the online databases of Science Direct, Pro-Quest and Web of Science, along with the search engine of Google Scholar. These keywords were chosen to cover a wide range of research investigating many aspects of transformation and sustainable development, such as leadership approaches, policy implications, educational practices, and incorporating SDGs into the HEI framework. These databases, widely recognized for their exhaustive coverage of academic literature in higher education and sustainability research, were utilized in this SLR. Prior education research has extensively used these databases [12,54–56,58]. By utilizing these reputable sources, we aimed to incorporate the field’s latest and most pertinent studies, following the principles and methods of SLR. The study selection criteria guarantee that the review is targeted and relevant. Inclusion criteria involve studies with the following characteristics:

- Focus on transforming HEIs for sustainable development.

- Investigate the role of leadership, management, and governance in promoting sustainability within HEIs.
- Discuss how SDGs, particularly SDG4, can be integrated into the pedagogical frameworks of HEIs.
- Examine how transformation and sustainability affect teaching, learning, and graduate employability.
- Are published in peer-reviewed journals, conference proceedings, or credible institutional reports.

Exclusion criteria are used to exclude research where

- Only primary or secondary education is covered, which is irrelevant to HEIs.
- Has no clear emphasis on sustainability or change in higher education.
- Are not based on actual study and do not provide significant insights into the subject.

We assessed the quality of the papers included in our analysis by considering many essential factors to ensure the strength and reliability of our results. The criteria encompassed research design, sample size, data-gathering techniques, and analytic strategies. While a formal quality evaluation tool was not utilized, these criteria provide a systematic method to evaluate the equivalence and reliability of the discovered publications. The studies were classified according to their design, which might be qualitative, quantitative, or a combination of both, to assess their methodological rigor and contextual relevance. Various research approaches, such as qualitative case studies, secondary research, and comprehensive survey designs, were identified in the study [59–61].

The research varied in sample sizes, ranging from small single-case studies to extensive surveys. Generally, higher sample sizes yielded conclusions that were more applicable to a broader population, whereas the smaller studies provided detailed insights but had limited applicability. The data-gathering techniques utilized in this study encompassed interviews, questionnaires, and secondary data analysis. To evaluate the reliability and effectiveness of these methods, an examination was conducted of the data collection process and its suitability for addressing the research topics at hand. In addition, the analytical approaches were evaluated to confirm their suitability for the data and study design. This included using statistical methods for quantitative studies and theme analysis for qualitative research.

The initial screening process involves searching the titles, abstracts, and bodies of publications using the database search windows. The search results were then organized in the order of relevancy. We initially evaluated the titles of articles and subsequently examined the keywords and abstracts to determine their relevance to our investigation. Following the first screening, we thoroughly examined the complete text of the publications to evaluate their methodological soundness and relevance. As a result, studies that did not match our quality standards or were methodologically flawed were excluded. The meticulous selection method guaranteed the inclusion of only rigorous research, establishing a solid basis for our narrative systematic literature review.

During the research evaluation, we discovered many methodological limitations, including possible biases arising from self-reported data and the influence of small sample sizes on the capacity to generalize the findings. The synthesis accounted for these limitations to ensure a fair and balanced interpretation of the findings. Research with methodological problems was acknowledged but assigned less weight in the overall analysis. This ensured that the more robust and trustworthy research had a more significant impact on our results. After identifying 562 data from Science Direct, Pro-Quest, Google Scholar, and Web of Science databases, 350 duplicates were eliminated. The Google search engine and citation searching identified an additional 18 documents, resulting in 230 articles. The titles, abstracts, and keywords were evaluated, and 29 records were excluded to limit the

scope to 201 research. After considering the document's relevance and methodological rigor, 22 research studies were deemed unsuitable and eliminated due to being out of scope and having limited rigor. This resulted in a final selection of 179 articles that matched the specific criteria for inclusion. The collaborative review approach involved independent appraisal of the included and undetermined articles by researchers, followed by a consensus to guarantee a fair and thorough evaluation of the findings. We ensured that our findings are founded on solid and dependable data by conducting a rigorous quality evaluation and selection process. This provides significant insights for scholars and practitioners in the area. We utilized the content and thematic analysis to identify patterns, categorize findings, and establish relationships between key concepts to analyze the selected studies systematically. The thematic analysis was conducted using the Braun and Clarke's six-step framework, which guaranteed a structured and reproducible data synthesis method [62]. At the outset, we familiarized ourselves with the data by meticulously reading and re-reading the selected studies, identifying recurring concepts related to data-driven leadership, employability, teaching strategies, and SDG integration in HEIs. The systematic identification of recurring terms and conceptual themes was enhanced through NVivo 14 software [63–65], which was preceded by rudimentary codes generated through initial manual coding. The identified identifiers were subsequently categorized into more general thematic categories, which enabled the identification of the most prevalent themes. An independent researcher reviewed the themes to guarantee accuracy and coherence, and any discrepancies were resolved through discussion. The final thematic structure comprised five critical themes: (1) the role of data-driven leadership in HEI transformation, (2) the impact of SDGs in higher education, (3) teaching, learning, and employability strategies, and (4) DEI in HEIs. These themes were explicitly defined to prevent conceptual overlap, guaranteeing a focused and structured interpretation of the findings.

A content analysis was conducted to quantify the presence and the relationships of the significant concepts across the studies, while a thematic analysis provided qualitative insights. Utilizing NVivo 14 software, we could ascertain the prominence of specific topics by extracting word frequency distributions, thematic co-occurrences, and conceptual linkages. For example, the term “graduate employability” was referenced in 65% of the studies. In comparison, “equity in HEIs” was discussed in 47% of the studies, suggesting substantial academic discourse on these subjects. In addition, using correlation matrices to map relationships revealed significant co-occurrences between themes such as “AI-driven education” and “predictive student analytics”, underscoring the emergence of emergent trends in technology-enhanced HEI leadership. The quantitative insights strengthened the study's reliability and profundity by complementing the thematic findings. By incorporating both the thematic and content analyses, we can guarantee a thorough and systematic synthesis of the existing literature, thereby facilitating a nuanced comprehension of the role of data-driven leadership in transforming higher education institutions.

3. Results and Discussions

3.1. Transforming Global HEIs Through Data-Driven Leadership

This narrative SLR guaranteed that the conceptual framework was actively employed to assess the evidence rather than functioning as a pre-existing bias toward data-driven leadership. The SLR presented a content analysis that contextualizes global higher education's transformative and sustainable development to classify how we can succeed in transforming HEIs. Today's education does not achieve its promise to assist in the formation of peaceful, just, and sustainable communities [66]. The present status of the world necessitates a fundamental shift in education to correct past injustices and strengthen our ability to collaborate for a more sustainable and equitable future [66]. Academic institu-

tions may promote the transition to a successful and sustainable future by fostering a new generation of citizens [67,68]. Education has been viewed as devoid of any direct purpose or external will, yet normative: it should assist in bringing society closer to an idealized humane future [5]—a future characterized by engaged and knowledgeable leadership, diversity, equity, and inclusion. Education can cause a fundamental shift in how we think, behave, and carry out our responsibilities toward one another [67].

Our study does not ignore the topic of stability in HEIs, as many academics have emphasized the importance of stability during the educational reform [69]. Maintaining a stable educational purpose is essential in a tumultuous environment [70,71]. While change is necessary, stability is critical to ensuring that academic institutions can adapt and survive in the face of change. To deal with the challenges of change, evaluating the status quo of HEIs can help [70–73].

Our stance is that educational transformation is essential for ensuring that HEIs remain dynamic, relevant, and effective in meeting the needs of students and society. It promotes innovation, equity, and lifelong learning, empowering individuals to make meaningful contributions to the world and address its pressing challenges. A long-term strategy that establishes a framework for stable, yet transformative education is required. Therefore, we argue that the solutions to some of the sustainable challenges of our time can be found in the transformation of the world's HEIs, and this can be accomplished through engaged and informed leadership, data-driven decision-making, diversity, equity, and inclusive education. Reviewing the literature, our positions in this study are discussed below in themes.

3.2. The Role of Data-Driven Leadership in HEI Transformation

A successful education transition depends on leadership that emphasizes transformative learning, organizational effectiveness, and innovation. However, the ongoing development in HEI environments requires integrating data-driven leadership into leadership and changing management methodologies [74–77]. As detailed by some scholars, datafication has far-reaching effects, and one of the most obvious is in the realm of education, where it not only alters the structure of classroom instruction but also influences how students of the future make sense of the world around them [28,61,78–80]. Data are collected at all levels of education (individual, classroom, school, state, and worldwide) and may pertain to all aspects of instruction and administration. As a result of this deluge of information, everybody involved in education—from policymakers and administrators to educators—must adjust how they think about and approach their work [61]. Knowledge-based decision-making at any level requires data that has been acquired, organized, analyzed, and commented upon—this procedure will improve students' academic outcomes by making the transition from one level of education to the next more accessible, both inside the school and throughout the broader education system [81].

A data-driven organizational baseline can only be established via the integration of technology, data, skills, and people inside HEIs if this transformation is to be a success. Educational leaders are empowered by technology, data, people, and expertise who exploit data and investigate how the analytics and insights derived from data are significant in empowering better leadership and informed decision-making about their students [82]. Hamilton et al. show that incorporating data into leadership choices significantly improves institutional performance by matching educational methods with market needs. This congruence results in enhanced student outcomes and institutional success [82]. Baker and Siemens present evidence that data analytics can uncover trends in student behavior, enabling educators to customize their teaching methods to accommodate various learning demands and improve educational effectiveness [83]. According to McCarthy et al., a

leadership team must be built on a digital transformation program that considers people, processes, technology, and data, with the data supporting and being integrated into every leadership decision and process [84]. HEI leadership will need to practice the art of investigation and listening with all academic stakeholders (education policymakers, management, teachers, institutional staff members, and students) and other stakeholders identified in the learning ecosystem [39] who can contribute beneficial outside perspectives that can start effective decision-making. Some studies have argued that the systemic growth of higher education is characterized by a gradual opening up to internal and external stakeholders and, consequently, an embrace of linked and open settings [51,85].

Ideally, this allows higher education institutions to adjust to the growing complexity of society and the changing demands of education by promoting interdisciplinary collaboration and co-creative problem-solving [51]. HEIs can effectively address the increasing diversity and promote fairness and inclusion through effective governance [86]. However, to attain excellence, HEIs must modernize their decision-making processes and implement data-driven governance models [1,32]. At all levels, data-driven decision-making has become fundamental in educational practice, influencing policy and institutional strategies [27,87]. Nevertheless, stakeholders must possess data literacy to convert insights into significant enhancements in higher education [88]. Faculty members may employ teaching, learning, and research data to improve student engagement and academic success. Conversely, they must also cultivate proficiency in data analysis and interpretation to optimize curriculum design, implementation, and evaluation [88]. Similarly, professional and support staff can enhance the students' learning experience, academic achievement, and employability by utilizing their data management and analysis skills to create comprehensive student profiles [88]. In HEIs, transformational leadership is indispensable for establishing support structures that facilitate institutional change and guarantee that HEIs are future-ready and adaptable.

According to Hamzah, universities must not only be reactive to an ever-changing external environment but also become proactive pioneers in providing innovative answers to regional and global concerns [1]. However, these leaders must be more aggressive in permitting some modification and redefining the schools to nurture frontier practices and create an urgency for change [89], as their responsibility is to guarantee that the institution has a suitable strategic plan and is efficiently implemented.

Several current concerns generate a discussion in HEIs on a worldwide scale, such as the employability of graduates, innovative teaching and learning strategies [39,90,91], inclusion, diversity, and equality–equity in HEIs [92,93]. The findings of many empirical studies by [94,95] reveal that data can be used to discover students at risk and understand the overall student population, as well as the learning environment and teaching processes. As many studies indicate, adding predictive analytics to institutional management information allows for better-informed decisions [24–26,95]. Identifying at-risk students enables universities, particularly program directors, to implement targeted intervention strategies to assist the students and improve success rates [95]. Using data analytics to identify students at risk early enables prompt interventions, enhancing student retention and success rates. Reference [96] presents empirical data demonstrating that using predictive analytics may detect students in danger of academic failure early on. This enables educational institutions to promptly implement targeted interventions, enhancing student retention and graduation rates. Research highlights the capacity of learning analytics to customize educational experiences, thereby improving student retention and achievement [97].

Data also help to reduce the gap between lower and higher-performing students and better prepare the teachers for guiding the students toward reaching their academic potential [23,94,98,99]. Furthermore, data mining can be used to understand the graduate level

of employability, offering opportunities and solutions for decision-makers to improve employability and propose relevant interventions [21,22,90,100]. Empirical studies emphasize the value of data collection and analysis in the educational ecosystem, assisting educators in making proactive decisions to enhance student performance and employability [101,102]. Educational data mining is valuable for studying academic data, identifying patterns, and supporting decision-making affecting graduate employability [103]. Moreover, the capacity of teachers to use data to establish the differences in learning provides timely opportunities for instructors to redirect or add resources to facilitate progression toward optimal teaching and learning patterns. This approach leads to a continuous, sustained improvement of pedagogical practice as data extract meaningful knowledge from large population sets [101,104,105]. Thus, data inform the decision-making process for university management and administration [101,103,104], transforming the entire education ecosystem.

Empirically, the link between data, HEI leadership, teaching and learning, graduate-level of employability, inclusion, and equity is a cyclical process. Data informs leadership decisions, which, in turn, influences teaching and learning practices [106,107]. The effectiveness of teaching and learning impacts the employability of graduates [39,90,108–110], and the focus on inclusion and equity ensures that all students have equal opportunities to succeed. By continuously monitoring and analyzing data, HEIs can improve their overall performance and contribute to the success of their students in the professional world while promoting a fair and inclusive educational environment [21,22,100,111]. To achieve an equitable and sustainable transition in universities, these concerns in academic contexts require attention, reevaluation of decisions, changes in policy, and remedies.

Figure 3 demonstrates the dynamic process propelling the transformation in HEIs, illustrating the cyclical and interdependent relationship between data, leadership, teaching and learning, and employability. Data informs leadership decisions by offering insights into future trends, student requirements, and institutional performance. Consequently, leadership creates innovative teaching and learning strategies per the institution's objectives, market demands, and inclusivity. These improved teaching and learning practices directly impact employability, providing graduates with the necessary skills to meet the needs of the labor market. Feedback from employability outcomes generates new data, which leads to the refinement of institutional strategies and the completion of the cycle. The feedback loops (blue-dotted arrow) illustrate the iterative nature of these relationships, in which leadership influences data collection and is informed by teaching outcomes. At the same time, employability orients teaching reforms and is enhanced by improved learning practices. This interconnected system emphasizes the alignment of HEI activities with sustainability, equity, and inclusivity objectives, guaranteeing ongoing adaptation and enhancement in response to changing educational and societal challenges.

With the support from the empirical studies above, we assert that data will be utilized to analyze and address key challenges in HEIs, enabling data-driven leadership to support the transformation. First, graduate employability depends on the combination of soft and hard talents, traits, competencies, subject-specific knowledge, and capacities that enable prospective university graduates to compete effectively in the labor market. Second, teaching and learning strategies play a crucial role in academic success. Teaching strategies refer to the methods used by lecturers to facilitate student learning. On the other hand, learning strategies encompass techniques, activities, and skills the students adopt to enhance their ability to acquire and retain course concepts, information, and background knowledge to meet specific academic objectives. Third, inclusion, equality, and diversity are tenets of HEIs. Educational diversity refers to the range of identities, orientations, demographics, distinctions, and views present in a group of individuals in an academic context. Equality in education signifies that all students have equal access to educational

resources and opportunities (sameness). Equity in education indicates that all academic environment participants receive the resources and opportunities they require based on their specific circumstances (fairness). Increasing everyone's access to education and their ability to participate in economic, political, and social life is the objective of inclusive education. It seeks to identify all barriers to education and include all students, regardless of their disabilities.

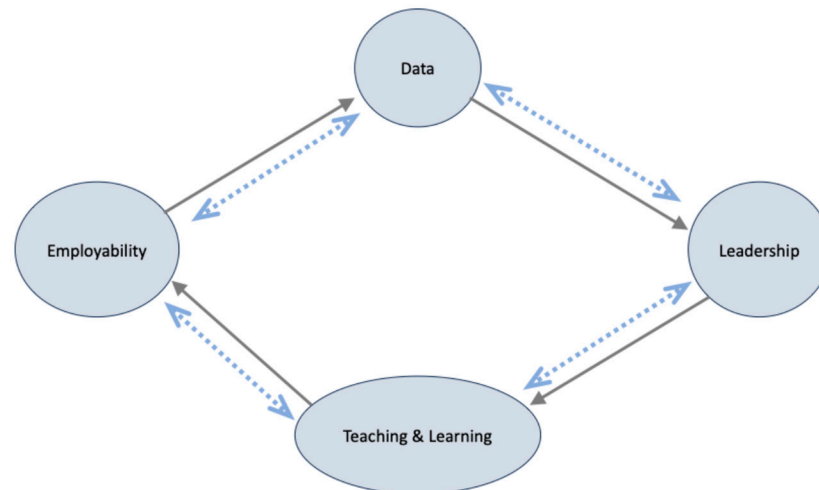


Figure 3. Interconnections between teaching, learning, and employability.

The issues of leadership, data, and the level of graduate employability are crucial in driving the success and effectiveness of HEIs and preparing graduates for the job market. We explore this link in detail below.

3.3. Data-Driven Leadership and SDG Integration in HEIs

As universities actively reflect on, discover, and advocate answers in conversation with society, sustainable development will be the primary framework for generating influence through university missions. Some scholars state that to achieve the necessary change in HEIs, sustainable development concepts must be at the center of their strategies [14,35]. HEIs must lead by example to influence the external society and sustainable development [12,27]. As a transformative agent, the higher education sector significantly contributes to a more sustainable and prosperous society [35,112–114]. In other words, HEIs are intrinsically obligated to make communities more sustainable by incorporating sustainability into their systems while considering their societal implications [12,13,115]. To build the quality of education in many parts of the world, a galvanized international scene has placed transformation at the top of the education agenda, but it requires leadership—a vision of change [116,117] that can directly shape the performances of their students, which in turn affects the market performance of the country [118].

Modern institutions of higher education that wish to prosper must look beyond teaching and learning and developing students' skills. Instead, they must establish enduring value engines that benefit all students. Students' growth, which is both transformational and long-lasting, is one of the essential variables that can assist in driving this ideal. According to UNESCO, learners must establish coherence between the world they encounter in school and the world they aim to construct outside of school [89], which extends to all other SDGs [66]. Consequently, HEIs will be crucial in achieving all other SDGs and transforming the world [12]. For example, a world where people are diverse, and it is acknowledged; a world where people are equal and experience equity and inclusion because of a transformation agenda that weaves in leadership that makes decisions based on

rich data collection, analysis, and interpretation of these data and proactively engages in data-driven transformational action.

We summarize the main results and arguments from the Results and Discussions section by giving a cohesive overview of the SLR's views on transforming global HEIs. This overview is also illustrated in Figure 4.

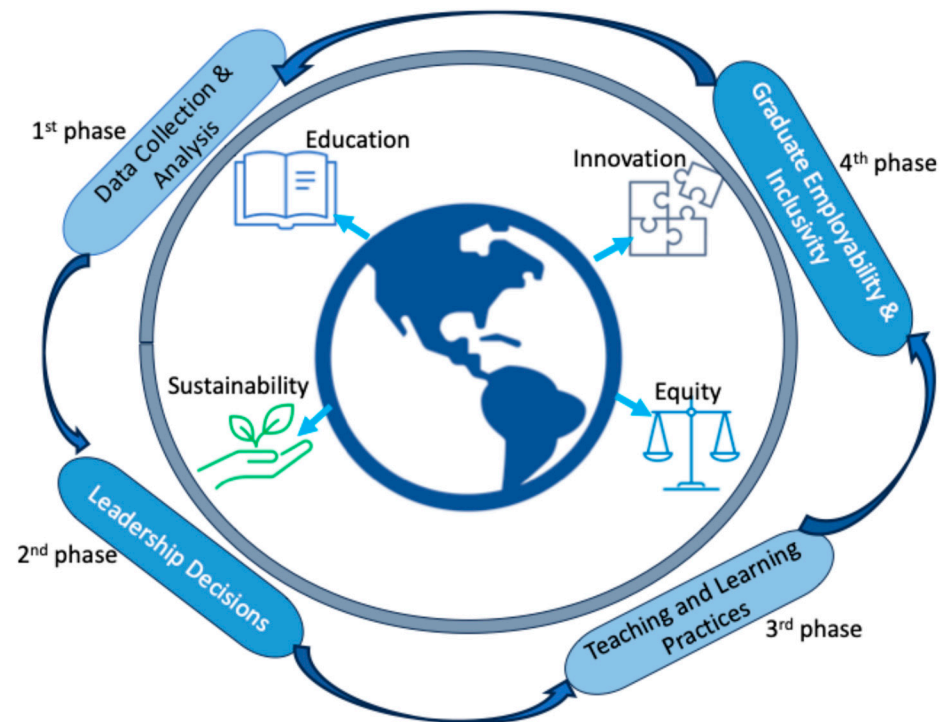


Figure 4. Transformation of global HEIs for sustainable development.

The analysis emphasizes the significance of maintaining stability during educational changes by balancing the necessity for change with preserving a steady educational objective. This balance guarantees that HEIs can adjust and persist in the face of transformative forces. The review promotes a comprehensive strategy for educational reform that combines data-driven leadership, diversity, equity, and inclusive education to address long-lasting difficulties.

Effective leadership is crucial for effective educational transitions. Incorporating data-driven leadership into change management approaches is essential for effectively dealing with the intricacies of HEI environments. As stated by Jarke and Breiter, datafication substantially influences educational systems and decision-making procedures [61]. The study emphasizes the importance of establishing a data-driven organizational foundation, which relies on technology, data, skills, and people, to enable effective leadership and well-informed decision-making. Effective governance is crucial for helping HEIs address growing diversity and advance equity and inclusivity. Transforming university decision-making and management is essential for achieving excellence through reformed governance. Stakeholders must be data literate to effectively interpret data and improve the quality and success of higher education. The study delves into the relationships between data, HEIs leadership, teaching and learning, graduate employability, inclusion, and equity. A circular process is recognized in which data guides leadership decisions, affecting teaching and learning practices [119] and subsequently influencing graduate employability and inclusion. HEIs can enhance their overall performance and support student achievement in the employment realm through continuous data monitoring and analysis.

3.4. Advancing Teaching and Learning Strategies with Data Analytics

The use of data in decision-making has grown widespread in HEIs [61,120]. The importance of data-driven decision-making has grown with accountability, school effectiveness, and transformative learning [25,26,82,121]. Effective teachers and schools use data to inform their educational methods and systems at all levels [121]. To address the problems posed by accountability measures, school leaders are urged to utilize data to influence teaching and learning methods [19,122,123] and to direct and monitor school reform [106]. The application of data-driven instructional leadership to enhance teachers' teaching practices enables instructors to reflect, collaborate, and improve instructional practice [61,107,124]. It is beneficial to comprehend the aspects of a school leadership that have a causal relationship with student outcomes and the leadership behaviors linked with excellent student outcomes [124]. To ensure DEI, however, including students as partners will empower students to take on more responsibility as students, and provide them with the opportunity to augment their voices and make meaningful contributions to HE spaces [125]. This inclusion aids in comprehending the problems and expectations of the students regarding essential teaching and learning concerns [125].

Indeed, transforming teaching and learning to improve student outcomes might involve many steps. Halverson et al. outlined how leaders can play an active role in using data to create structures that instill change in HEIs through data acquisition, data reflection, program alignment, program design, formative feedback, and test preparation [106]:

- Acquire data: Seek, collect, and organize instructional-relevant knowledge.
- Reflect on the data: Analyze student learning to set objectives for enhancing teaching and learning and arrange opportunities for teachers and leaders to make sense of data, define problems, and collectively decide goals and a course of action.
- Align the program: Ensure the school's instructional program is aligned with relevant subject and performance requirements and classroom content to increase student learning and fulfill their needs.
- Design or alter curriculum: Adapt pedagogies, student assistance programs, and instructional methodologies; schools can respond to perceived instructional needs by enhancing student learning.
- Format feedback structures: Develop learner-centered, iterative assessment cycles to generate continual, timely information flows that enhance student learning and instructional program quality across the school. Continuous feedback loops derived from data analysis provide valuable insights that enhance teaching techniques, ultimately improving the overall quality of education and students' employability. Studies emphasize the significance of ongoing feedback loops in guiding instructional practices, leading to enhanced educational quality and employability [52]. Chigbu et al. discovered that using data analytics improves learning outcomes and increases the employability of graduates by aligning educational programs with the market's demands [39]. Structures for formative feedback contain information about student learning or instructor practice.
- Test preparation: Implement strategies to inspire the students and establish methods for enhancing their performance on state and district examinations.

Data-driven transformations occur when the data enable HEIs to uncover challenges and possibilities in their teaching and learning techniques that they may not have been aware of otherwise [52,126,127]; nevertheless, institutions must base their judgments on solid and reliable data [59]. Measuring a learner's route to competency provides detailed data in areas where genuine misunderstandings may occur, which requires creativity and a significant change in data-driven processes [19,52,122,127]. It necessitates replacing the linear, content-first approach with a formative assessment approaches with specific

indicators. It also entails transitioning from a conventional, designed-for-the-average-learner experience to a unique, individualized experience.

The study of data can offer an overview of what students know, what they should know, and what can be performed to suit their academic requirements with the proper analysis and interpretation. No single evaluation can provide educators with all the information they need to make well-informed instructional decisions; thus, it is prudent to utilize multiple data sources. Schools gather voluminous data on students' attendance, conduct, performance, and administrative and perceptual data gleaned through surveys and focus groups. However, when it comes to enhancing education and learning, it is not the quantity of data that matters but how it is analyzed and utilized. By leveraging data to inform decision-making, educational leaders can develop and implement teaching and learning strategies that cater to the unique needs of their students and align with broader institutional goals. Some studies show that utilizing educational data can customize learning, enhance academic outcomes, and improve teachers' skills and self-confidence, particularly when accompanied by professional development and integrated into teacher training programs [128–130]. Data-driven decision-making enables a culture of continuous improvement, ultimately leading to equality and inclusion in educational settings. The link between data, diversity, equality–equity, and inclusion in global HEIs is critical to creating a supportive and enriching learning environment for all students and fostering a sense of belonging within the institution.

3.5. Employability Through Data-Driven Skills Development

Employability and job preparedness are increasingly crucial in the global labor market and higher education environment [22,90,100,131–133]. This has increased calls for education and training to have a worldwide perspective [134]. Assessment of learning outcomes has always been an internal affair of HEIs; however, conventional collegial procedures are no longer adequate. In analyzing the results of higher education students, such as employability, policymakers face a significant knowledge and data deficit [23–25,134]. Using data, HEIs may determine the global best practice employability models and how career services and academic departments collaborate [25,28,79,82,135]. Universities might seek to integrate careers and job support into curriculum creation and the implementation of broader pedagogical practice in HEIs [136]. In addition, ref. [137] argues that incorporating global perspectives into any engineering, science, humanities, or business-related course can address the lack of employability skills among higher education graduates [136].

The supply of adequate and coherent data, the identification of the most relevant approaches, and the availability of institutional resources and knowledge are only a few of the difficulties inherent in creating systems to predict skill requirements [98,137,138]. Through this review, we argue that HEIs should be actively involved in the skills analysis, skills identification or matching, and skills training of their students and should ask questions such as the following:

- What skills are required for employment in general? What skills do our students possess, and where do they need advancement?
- What skills do our learners require for their desired careers?
- How can we, as HEIs, provide our students with the opportunity, training, or exposure to equip them with the needed knowledge and skills?

The skills gap is an urgent policy and transformation issue; HEI stakeholders and leaders are at the forefront of this conflict. To make competent and well-informed decisions, it is crucial to comprehend the difficulties surrounding data-driven skills development in higher education. The phrase “skills gap” refers to a fundamental mismatch between the abilities required by the labor market and those possessed by university graduates. Due

to this mismatch in a globalized society, many people's career goals are limited to a small number of work opportunities. This mismatch makes it difficult for graduates to obtain employment [139].

The leadership of HEIs is essential to resolving this gap and mismatch, and data may motivate this leadership to be inquisitive (asking what, when, why, and how), informed, proactive, engaged, inclusive, efficient, and transformational. Various data sources are available to identify significant historical and present patterns in the demand for (economic activity) and the supply of skills (education characteristics). In this context, "data" refer to the government's occupation list data, a country's central statistical agency, the skills gap and mismatch report and data, public employment services, LinkedIn's scarce skill data, the labor force survey, and advertising data from internet and print media. Moreover, these data sources gather and publish information identifying sectors and occupations with skills shortages and why some openings are difficult to fill. Therefore, studies should ideally consider various data and indicators to obtain a comprehensive and objective picture [137]. The initial phase of the procedure is to collect data to identify problem areas or the possibility of their emergence. HEIs must analyze the essential skills they require of their students and the talents they already possess and compare them to what the labor market needs (skills gap analysis). This sort of study may give leaders of HEIs knowledge, allowing them to take additional steps to boost the employability of their students. These efforts will include the following:

- Re-examining the teaching and learning practices that influence the students' skill development to determine if they are successful and influential.
- Encourage students to be lifelong learners and adapt to different educational opportunities (such as micro-credentials, skills certifications, and short courses).
- Create degrees and build curricula that are driven by the abilities desired by the labor market (moving away from repairing current courses to the ongoing revamping of courses [what does the labor market currently need?]).
- Integrate experiential and project-based learning into the core of the student experience.
- Create entrepreneurial channels and ecosystems for students.
- Establish strong partnerships with other, more prominent HEIs, corporations, and economic developers who can provide insight on overcoming the mismatch between students and jobs.
- Promote work experience, apprenticeships, and fellowships for students.

HEIs have always discovered innovative methods to prepare the next generation of students for future occupations, and this generation will be no exception [39]. However, due to rapid technological change, the Fourth Industrial Revolution (4IR), continued automation, ecological change, and the demand for sustainable development, higher education will need to make more efforts to transform the economy and society through the sustainable skill development of incoming graduates before entering the workforce [20,35,140–142]. According to some academics, many businesses are unwilling to train graduates [143], preferring instead to hire recent graduates from academic institutions that they feel have adequately prepared their graduates for the workforce. Additionally, the graduate employment rate is frequently used to evaluate the quality of university provisions [144–147]. The issue is that most employers equate prestige with ability, and perceived university quality is a primary determinant of which applicants from which universities are recruited—they only recruit at universities that they believe have prepared their students to be work-ready, which saves time and money for the employers [148]. By doing that, biases, exclusion, and unfairness are created. Notably, this method adopted by several companies has increased the graduate employment rate at certain HEIs and stalled it at many others. This is where the student's chosen HEI fails the student after the student has invested time and resources;

graduates are reduced to involuntary part-time work, which is now unsustainable for the graduates.

In summary, we agree that effective leadership in HEIs relies on data-driven decision-making processes that align educational offerings with the demands of the job market [60,149]. For instance, some studies prove that continuous data analysis and adaptation in HEIs enable leaders to respond to changing market needs through strategic planning, advanced analytics, and innovative teaching and learning strategies [150,151]. The link between leadership, data, and teaching and learning strategy is fundamental to effective educational institutions. Leadership plays a pivotal role in shaping the direction and vision of an academic institution, and data-driven decision-making is crucial for informing and improving teaching and learning strategies which we explored.

3.6. *Promoting Diversity, Equity, and Inclusion (DEI) in HEIs*

Inclusive and transformational education must guarantee that all students in all nations have unfettered access to and participation in education that is sustainable for them during and beyond graduation. Students at HEIs are more varied than ever, comprising numerous races, nationalities, socioeconomic backgrounds, languages, ethnicities, and many religious views, disabilities, and personalities [152–154]. Due to several major global trends, such as demographic shifts, migration, refugee crises, rising inequalities, climate change, gender identity and sexual orientation, special education needs, and giftedness, education policymakers are increasingly concerned with equity and inclusion. These changes have contributed to the rising variety of country populations and have raised questions about the capacity of educational institutions to be inclusive of all students [86].

There is a need to address DEI concerns in higher education contexts so that educational leaders may exemplify the stated ideals of their institutions while working to improve HEIs [155]. Diversity and inclusion in higher education are essential for a quality university education that prepares students for life and work. Inclusion is a problematic aspect of DEI to define and quantify, which is understandable [156]. However, it is essential to recognize that students are less successful when they do not feel involved or engaged in their college experience [90,143,156]. According to the Higher Education Diversity, Equity, and Inclusion Survey, most students say their school takes DEI concerns seriously, although many believe there is space for improvement [157]. Under these conditions, HEIs must develop ways to prioritize student voices in DEI work [14,158] through various tools, such as surveys, online polls, focus groups, committee representation, and other strategies to discover students' needs and gaps [156]. Unfortunately, little empirical data supports complete inclusion for all students and even less on the significance of data-driven decision-making in inclusive education [15].

Consequently, with data utilization, HEIs may progress more on their DEI journey. Universities that are willing to conduct a rigorous analysis of themselves as an academic body, to identify the gaps honestly, to promote empathetic listening to marginalized groups and stakeholders, and to implement interconnected, university-wide systemic changes will be in a much better position to achieve sustainable DEI. Consequently, they will develop an academic community that is better and more caring for everyone, not just under-represented groups. Moreover, educational leaders can evaluate whether initiatives and policies have favorable or harmful effects on students based on race, age, origin, learning milestones, disability, etc. This crucial information enables educational leaders to execute the most promising initiatives and push for legislation that removes exclusion, inequality, learning, and skills gaps most effectively.

Data-driven decision-making is a lifelong commitment to a philosophical, comprehensive shift toward continual improvement [121]. Higher education is crucial in attaining

SDG4 on education: inclusiveness, equity, and excellent education for everyone [159]. To be an inclusive accelerator of innovative societies, however, higher education must collaborate with governments and development agencies to solve difficulties such as DEI [160]. Inclusive practices must be a shared duty among educational community members to facilitate the full development of all members [160].

Components of responsible inclusiveness in education include full inclusion, putting the students' needs first, and re-evaluating the effectiveness of curriculum and instructional practices to meet the needs of diverse students [15,39,161]. We must address DEI in the classroom (representation of all students) and the curriculum, but also in campus buildings, distant experiences, and extracurricular activities [86,161]. The curriculum is vital to enabling a change toward a comprehensive and ecosystem-based approach to career and employability offerings [135].

Evidence-based practices and data-based decision-making within inclusive education prioritize satisfying the requirements of diverse students, such as those with reading difficulties and intellectual disabilities [15,161,162]. While data are an essential element of DEI, decision-makers must evaluate it to guide judgments about how to successfully support students by integrating it with pedagogical and subject expertise and translating it into a helpful action plan considering the context [15]. While teachers routinely collect data on student progress, they are less likely to use it to inform their teaching strategies or programming decisions. Therefore, teachers should receive the training necessary to interpret student data [15,163,164].

It is well-known that many HEIs focus data on enrollment, test scores, grades, and graduation rates as the key performance indicators. However, research indicates that HEIs employ multiple criteria to evaluate performance beyond these traditional metrics. These additional indicators include statistical measures, predictive indicators, soft skills, and success in professional exams [165–167]. In contrast, diagnostic data on their equity ecosystems is often overlooked. If data enable HEIs to provide 'access' to students who would not otherwise have it, this is a step toward equality. Still, it is not sufficient to establish equity. Having a varied student body (poor and high economic backgrounds, race, nationality, gender, ethnicity, sexual orientations, disabilities, religion, etc.) in higher education and its disciplines is a first step. Individuals should be encouraged, based on their individuality, to experience a sense of belonging once these diverse student groups are in institutions and classrooms. This is where data-driven HEI transformational leadership comes in, which entails gathering, analyzing, and interpreting each group of students, simultaneously revealing their individuality and necessitating equitable action plans.

Transformation of HEIs through equity requires the active participation of all student groups in decision-making. It is optimal to have effective decision-making leadership when those affected or targeted groups contribute their voices to the decision that will transform their current circumstances. Students with poor socioeconomic positions, students with learning difficulties, minority students, and students from unstable households—referred to as "at-risk students"—are especially vulnerable to losing their voices during creating and implementing strategic plans. In other words, transformational leaders in the education setting evaluate the lived experiences of all the students they instruct and promote change that improves those experiences for everyone. As a component of inclusive, sustainable, and transformative development, we can argue that all stakeholders (such as teachers, sponsors, management, administrators, service providers, policymakers, and students) in HEIs must become aware of and comprehend equity issues in academic settings. Transformation and sustainable growth are unlikely to occur in any HEI setting without (1) recognizing, appreciating, and valuing diversity; (2) treating everyone fairly for equity; and (3) engaging, embedding, and developing inclusiveness.

Transforming HEIs for sustainable development [168] is essential for preparing future generations, fostering responsible citizenship, advancing research and innovation, contributing to global goals, and making meaningful contributions to a more sustainable and equitable world. As hubs of knowledge and learning, universities have a significant role in driving positive change and championing sustainability efforts.

3.7. Policy Implications and Institutional Recommendations

Data-driven leadership is crucial for formulating effective policies and governance structures in HEIs, ensuring these institutions are robust, flexible, and aligned with SDGs. Table 1 summarizes that the results of this SLR demonstrate that incorporating data-driven decision-making into institutional leadership improves educational efficacy, accountability, and innovation [27,169]. The subsequent sections delineate the principal policy implications and recommendations for HEIs and policymakers derived from these findings.

Table 1. Implications of SLR on HEI transformation.

Aspect	Implications of SLR
Data-Driven Leadership	Highlights the necessity of integrating data-driven decision-making into leadership and governance structures to enhance institutional effectiveness, accountability, and innovation. Emphasizes the importance of a data-driven organizational baseline to support informed leadership and sustainable development [170].
Stability vs. Change	Stresses the importance of balancing institutional stability with the need for transformative change to ensure HEIs remain adaptable, resilient, and aligned with sustainability objectives. Advocates for structured change management approaches that leverage data analytics for strategic planning.
Holistic Transformation	Promotes a comprehensive approach to the HEI transformation by integrating sustainability principles, digital transformation, and inclusive education into institutional policies. A data-driven culture fosters innovation and long-term educational sustainability [171].
Empowered Leadership	Identifies data-literate, proactive, and engaged leadership as key to HEI success. Recommends professional development programs to enhance faculty and administrative capacity for utilizing data analytics in strategic decision-making and institutional growth [170,172].
Educational Practices	Highlights the need to embed sustainability principles into curricula, ensuring students develop data-driven problem-solving skills and are prepared for the evolving job markets [173]. Encourages the use of AI and predictive analytics to personalize learning experiences and improve academic outcomes.
Interconnections	Explores the interdependent relationship between data, leadership, teaching, learning, employability, and inclusivity. Suggests cross-sector collaborations between HEIs, businesses, and governments to enhance educational relevance and graduate employability.
Long-Term Strategy	Posits for a structured, long-term vision in HEIs centered on data-informed decision-making, digital transformation, and sustainability. Recommends continuous monitoring and evaluation of institutional progress toward global education and sustainability benchmarks.

3.7.1. Implications for HEIs and Policymakers

Table 1 illustrates the primary implications of data-driven governance for transforming HEIs, emphasizing critical areas such as long-term strategy development, holistic transformation, empowered leadership, and stability versus change. According to the results, HEIs must establish governance structures that promote evidence-based policymaking and consistently incorporate data into decision-making and leadership [87,174]. To cultivate a

data-driven institutional culture adaptable to new challenges, institutions should integrate sustainability principles into their operational strategies and curricula [175].

HEIs should foster collaboration with industry, government, and community organizations to enhance institutional effectiveness, ensuring that educational offerings meet sustainability objectives and labor market demands [176]. Furthermore, professional development programs must prioritize data literacy among faculty and staff, providing educators with the knowledge and abilities to make data-informed decisions [172]. Additionally, institutions must implement digital transformation strategies that improve administrative efficiency, student engagement, and teaching innovations, guaranteeing that HEIs can effectively leverage technology for educational advancement [169].

For policymakers, Table 1 emphasizes the significance of establishing transparent regulatory frameworks that encourage data-driven governance models in HEIs [27,177]. It is imperative to offer financial incentives to institutions dedicated to sustainable transformation, particularly in the funding associated with faculty training, technology adoption, and capacity-building. In addition, to facilitate the adoption of the best practices in data governance, sustainability, and digital transformation, policymakers should encourage the exchange of knowledge at national and international levels [170,178]. Moreover, institutional accountability necessitates HEIs monitoring and reporting on their sustainability initiatives and data-driven policies.

3.7.2. Recommendations for HEI and Policymakers

This study offers strategic recommendations for HEIs and policymakers to facilitate sustainable institutional transformation and enhance data-driven leadership, as informed by Table 1 and the SLR findings.

Data analytics, digital transformation, and evidence-based decision-making are indispensable components of a comprehensive sustainability strategy HEIs must establish [175]. In addition, institutions must incorporate sustainability principles into curricula across all disciplines, providing students with environmental literacy and data-driven problem-solving abilities. Holter and Frabutt propose that faculty training programs should be broadened to focus on data literacy, AI integration, and predictive analytics in education, preparing educators to utilize technology to enhance learning outcomes [172]. Additionally, HEIs must increase its investment in the institutional infrastructure by implementing sustainable, energy-efficient campus solutions that reduce their environmental impact [179]. Encouraging cross-sector collaborations with industry leaders, governments, and civil society is crucial for bridging the divide between academic research, employability, and sustainability policies [178].

To facilitate transformation in HEIs, policymakers must allocate funding for technology-driven innovations and sustainability-focused education research [171]. To guarantee accountability and transparency, policymakers should also require HEIs to submit sustainability performance reports [177]. Facilitating international knowledge-sharing platforms will allow HEIs to exchange expertise in implementing sustainability and data-driven governance. In addition, integrating sustainability benchmarks into accreditation requirements will reinforce the dedication to the highest standards of digital transformation and sustainability [178]. In conclusion, governments should support green campus initiatives, including renewable energy projects, green building certifications, and carbon footprint reductions, to ensure that HEIs actively promote environmental sustainability.

3.7.3. Limitations of the Study and Future Research

This SLR has offered valuable insights into the impact of data-driven leadership on the transformation of HEIs towards sustainable development. However, it is essential to acknowledge several limitations to contextualize the findings and inform future research.

First, the reviewed literature may contain selection bias, as the study primarily relied on published research accessible in specific databases. This limitation may limit the inclusion of regional studies, institutional data, and unpublished reports that could provide alternative perspectives on data-driven leadership in HEIs [170]. Furthermore, this investigation's lack of longitudinal data limits its ability to assess the long-term repercussions of data-driven decision-making on institutional transformation, sustainability, and student outcomes [169]. Future research should incorporate longitudinal methodologies to track the development of data-driven strategies adopted by HEIs [171].

Secondly, although this investigation emphasizes the benefits of data-driven leadership, assessing alternative perspectives is crucial. Some scholars warn that an excessive reliance on data analytics may lead to a rigid, metric-driven approach that neglects qualitative insights, including student experiences, faculty perspectives, and institutional culture [170]. Additionally, ethical concerns regarding algorithmic bias, surveillance, data privacy, and misinterpretation of learning analytics remain substantial [173]. To guarantee equitable and ethical data governance practices in HEIs, future research should examine the equilibrium between data-driven decision-making and human-centric leadership [174].

Third, it is imperative to conduct further research on institutional resistance to data-driven transformation. Faculty and administrators may be apprehensive that excessive reliance on data could compromise academic autonomy, result in distrust in data interpretation methods, or underscore a lack of digital literacy [175]. Research should examine the barriers to adoption and develop strategies for fostering a culture of data-informed decision-making and promoting data literacy in HEIs [175]. Comparative studies between institutional size, funding, and regional policies could provide valuable insights into the impact of data-driven leadership on the efficacy of larger, well-funded universities and smaller, resource-constrained institutions.

Finally, although the primary focus of this review is the intersection of data-driven leadership and SDGs, it is imperative to conduct further research to investigate the interplay between various sustainability initiatives, such as those involved in climate action, equity education, and global knowledge-sharing [176]. Future research should assess the environmental, social, and economic consequences of sustainability initiatives in HEIs, considering their scalability across various academic contexts. Furthermore, research on student engagement in sustainability initiatives and the role of student-led advocacy in promoting institutional transformation could provide valuable insights into HEI leadership models that effectively balance data, governance, and student agency [172].

4. Conclusions

Data-driven leadership is no longer a theoretical possibility but is instead necessary for transforming HEIs. To remain relevant, inclusive, and sustainable, HEIs must incorporate data analytics into their institutional strategy, governance, and decision-making as digitalization transforms the global education landscapes. This investigation has investigated the function of data-driven leadership in transforming HEIs towards sustainable development. By integrating data analytics into governance, leadership, and decision-making processes, institutions can improve students' employability, institutional efficiency, and teaching and learning strategies while promoting DEI. However, our findings indicate that, despite the widespread recognition of the advantages of data-driven leadership, its successful implementation is contingent upon institutional fitness, stakeholder buy-in, and a balanced

approach to data-driven and human-centric decision-making. This study's primary contribution is synthesizing empirical research and theoretical perspectives, establishing a structured framework for HEIs to implement data-driven governance consistent with the SDGs. This review delineates specific mechanisms through which institutions can systematically incorporate data analytics to drive meaningful change, although some aspects of data-driven transformation may seem intuitive. The study also underlines the ethical and operational obstacles associated with data-driven decision-making, underscoring the necessity of institutional policies that balance academic integrity, inclusivity, and technological efficiency. Although this research provides valuable insights, it is crucial to recognize three significant limitations. First, the under-representation of institutional reports and regional studies may introduce selection bias due to the reliance on published research. Thus, future research should integrate primary institutional data, case studies, and gray literature to gain a more comprehensive knowledge. Secondly, the dearth of longitudinal studies restricts the capacity to evaluate the long-term effects of data-driven leadership on the sustainability of HEIs and student outcomes. Consequently, future research should monitor institutional transformations over time to validate these strategies empirically. Third, this review emphasizes the advantages of data-driven decision-making; however, faculty resistance, algorithmic bias, and data privacy are not adequately addressed. Therefore, future research should investigate the ethical implications of AI-driven learning analytics and examine strategies for navigating institutional resistance to digital transformation.

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